For Medical And Environment Sensing Applications

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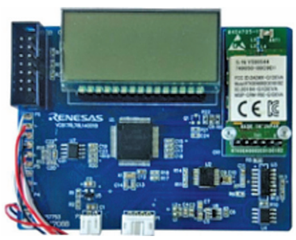
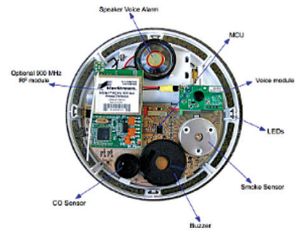
Reference Design for Handheld TDS Meter	Reference Design for Smart Body Fat Scale	Reference Design for CO Detector
		
Reference design for a handheld TDS (total dissolved solids) meter that employs direct measurement of conductivity of the solution to detect the TDS.	Reference design for a smart body fat scale that can display body weight, body mass index (BMI), body fat percentage, and body impedance.	Reference design for a carbon monoxide (CO) sensor that can measure amount of CO and smoke particles in the atmosphere and alert the users when the level of CO particles becomes dangerous.
CN0411 from Analog Devices (ADI) is a reference design for a compact total dissolved solids (TDS) measurement system. The TDS monitor uses the direct measurement of the conductivity of the solution technique to detect TDS. The main advantage of this design is that it uses a standard BNC probe connection, making it both faster and cheaper to design. The frequency of the excitation in this design is controlled by a PWM signal from the microcontroller. The reference design is also Arduino form-factor compatible. For higher accuracy, the reference design also has a temperature compensation that is performed using either a 100Ω or 1,000Ω 2-wire RTD. The design is capable of measuring low to high conductivity levels ranging from 1μs to 0.1s and can accommodate 2-wire conductivity probes of different cell constants from 0.01 to 10.	This Smart Body Fat Scale from Renesas is a reference design capable of measuring weight and displaying complete information about your body composition. The reference design includes four weighing sensors along with four metal electrodes for body impedance measurement. The reference design employs a Renesas RL78/L1A microcontroller (MCU) that adds Bluetooth connectivity to the device. The user needs to feed information such as height, gender, and age, while the weight and body impedance is measured by the smart scale. The MCU utilises all the above data to compute body fat percentage and other health parameters, which are displayed on an LCD monitor. Furthermore, the data is displayed on the smartphone application simultaneously. The smart body fat scale is suitable for healthcare bio-sensing applications.	This Carbon Monoxide (CO) Detector reference design from Zilog is based on electrochemical-cell sensor technology. The electrochemical sensors are more robust than any other CO detector. It is based on a Zilog's Z8 Encore XP microcontroller (MCU) and the Sixth Sense ECO-Sure (2e) CO sensor. The on-chip integrated hardware of the MCU minimises the board space with very few external components. The MCU also implements a power management routine to conserve power. An optional 900MHz RF module can be added to the reference design that can add wireless networking and monitoring features, making it suitable for various applications. This unit runs on three AA-size batteries and has a blinking LED indicator to indicate the status of the detector. With little modification, the design can be run using rechargeable lithium-ion batteries.
The reference design can be used for measuring the total dissolved solids (TDS) in water.	The reference design is suitable for measuring body composition.	The reference design is suitable for use as a CO sensor used in refineries, garages, homes, and waste-water treatment systems to make them safer.
Analog Devices (ADI)	Renesas	Zilog
https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-handheld-tds-metre	https://www.electronicsforu.com/news/reference-design-for-a-smart-body-fat-scale	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-co-detector